COLLECTING DATA IS NOT ENOUGH

We must integrate evidence to support timely decisions.

The value of surveillance lies not in the number of signals reaching decision-makers, but in the ability to interpret them and act in time.

THE TERRITORY ALREADY HAS SENTINELS

Wildlife, livestock and aquatic species may reveal exposures requiring targeted investigation.

A sentinel does not establish a diagnosis: it indicates where to investigate. An observed signal initiates verification; it does not complete a risk assessment.

THE ENVIRONMENT DOES NOT END AT THE FRONT DOOR

Dogs and cats share indoor air, dust and surfaces with people.

Shared environments do not imply identical doses. Companion animals can provide biological signals of overlooked household exposures, but these signals require species-specific interpretation.

ONE COMMON DIRECTION

Environmental science, veterinary medicine, toxicology and epidemiology must converge.

Environmental monitoring identifies sources and pressures; veterinary medicine contributes sentinel evidence and food-chain knowledge; toxicology interprets exposure and effects; epidemiology examines population-level outcomes. Joint interpretation can support proportionate decisions.

AN OPPORTUNITY FOR INTEGRATION

Not only what is present in water, but also what biological signals it produces.

Directive (EU) 2026/805 addresses water pollutants and a pilot approach to effect-based monitoring. Directive (EU) 2024/3019 concerns urban wastewater and public-health surveillance. Effect-based methods (EBM), wastewater-based epidemiology (WBE) and antimicrobial resistance (AMR) surveillance are distinct but complementary.

ONE PILOT BASIN, CLEAR RESPONSIBILITIES

Detect → Verify → Interpret → Decide

Remote sensing, drones and sensors can detect anomalies. Sampling, chemistry and biological methods can verify them. Environmental, veterinary and public-health expertise can interpret findings. Competent authorities retain responsibility for proportionate action. Each alert needs an owner, verification pathway and response timeframe.

TIME SAVED IS MEASURABLE PREVENTION

Technology detects. Expertise interprets. Institutions act.

The goal is to shorten the interval between the first reliable signal and a justified intervention.

Everything is in plain sight. What is missing is knowing how to see it.